

ConstantPad2d#

<https://pytorch.org/docs/stable/generated/torch.nn.ConstantPad2d.html>

Description:

Pads the input tensor boundaries with a constant value. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses $(padding_left, padding_right, padding_top, padding_bottom)$. Input: (N, C, H_{in}, W_{in}) or (C, H_{in}, W_{in}) . Output: (N, C, H_{out}, W_{out}) or (C, H_{out}, W_{out}) , where

Function Signature

```
class torch.nn.ConstantPad2d(padding, value)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ConstantPad2d(2, 3.5)
>>> input = torch.randn(1, 2, 2)
>>> input
tensor([[[ 1.6585,  0.4320],
          [-0.8701, -0.4649]]])
>>> m(input)
tensor([[[[ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  1.6585,  0.4320,  3.5000,  3.5000],
          [ 3.5000,  3.5000, -0.8701, -0.4649,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000,
3.5000]]]])
>>> # using different paddings for different sides
>>> m = nn.ConstantPad2d((3, 0, 2, 1), 3.5)
>>> m(input)
tensor([[[[ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000],
          [ 3.5000,  3.5000,  3.5000,  1.6585,  0.4320],
          [ 3.5000,  3.5000,  3.5000, -0.8701, -0.4649],
          [ 3.5000,  3.5000,  3.5000,  3.5000,  3.5000]]]])
```

Detailed Documentation

Documentation

Pads the input tensor boundaries with a constant value.

For N-dimensional padding, use `torch.nn.functional.pad()`.

`padding` (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 4-tuple, uses $(padding_left, padding_right, padding_top, padding_bottom)$

Input: (N, C, H_{in}, W_{in}) or (C, H_{in}, W_{in})

Output: (N, C, H_{out}, W_{out}) or (C, H_{out}, W_{out}) , where

$H_{out} = H_{in} + padding_top + padding_bottom$

$W_{out} = W_{in} + padding_left + padding_right$